

The code for `$a_2$` is:

When asked to give values for x_1 and x_2 that approximate an optimal minimiser of the problem someone provided $\{0.4, 0.6\}$ as a solution. Is this at most 0.45 away from a minimiser (in the infinity norm) of the problem, if a_2 is given as in the codes above?

Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Yes.

Explanation:

Evaluating the limit gives $a_2 = 1 + \varepsilon$, where $\varepsilon \approx 2 \cdot 72^{-10^{24}}$ is an astronomically small positive number.

positive number.

The optimisation problem is to minimise $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$.

Since $a_2 > 1$, the unique minimiser is $(x_1^*, x_2^*) = (0, 1/a_2) \approx (0, 1)$.

The infinity-norm distance from the provided point $(0.4, 0.6)$ to $(0, 1/a_2)$ is

$\max(|0.4 - 0|, |0.6 - 1/a_2|) = \max(0.4, 0.4 - \varepsilon') = 0.4$ (where $\varepsilon' \approx \varepsilon$).

Since $0.4 \leq 0.45$, the provided solution is at most 0.45 away from the true minimiser.